

Literature Review Memo

Energy-Aware Expert Routing in Mixture-of-Experts Models
M1 Deliverable — Hrishabh Gupta — April 2026

1. Introduction & Scope

This memo synthesizes findings from 15 core papers spanning three research pillars: Mixture-of-Experts (MoE) architecture and routing (7 papers), MoE behavioral analysis (3 papers), and Green AI / energy measurement (5 papers + nvidia-smi documentation). The review was conducted using a triage reading approach (abstract/conclusion first, then methods, full read if highly relevant) to manage volume while maintaining depth on foundational works.

The primary goal is to identify the research gap that our project fills and to document baseline approaches for comparison. Our project proposes a novel energy-aware routing mechanism for MoE models: augmenting the standard top-k gating function with a real-time energy penalty term $\lambda \cdot e_{\text{gpu}}$, derived from per-GPU power states via pynvml.

2. MoE Routing: State of the Art

The MoE routing paradigm was established by Shazeer et al. (2017), who introduced the sparsely-gated MoE layer with noisy top-k gating: $\mathbf{G}(\mathbf{x}) = \text{Softmax}(\text{TopK}(\mathbf{H}(\mathbf{x}), \mathbf{k}))$. Two auxiliary losses (importance and load) prevent expert collapse. This was simplified by Fedus et al. (2022) in the Switch Transformer to k=1 routing with a single balance loss, achieving 7x pre-training speedup on C4. Lepikhin et al. (2020, GShard) used top-2 routing to scale to 600B parameters across 2048 TPUs for multilingual MT.

More recent work explores routing alternatives. Riquelme et al. (2021, V-MoE) introduced Batch Priority Routing for adaptive per-image compute in vision, finding that experts tend to learn similar functions (low diversity). Puigcerver et al. (2023) proposed Soft MoE, replacing hard routing with fully differentiable weighted combinations, eliminating token dropping and expert collapse entirely. Mustafa et al. (2022, LIMoE) demonstrated multi-objective routing (accuracy + per-modality entropy regularization) for multimodal tasks. Kool et al. (2024, MoE-Mamba) showed MoE routing is architecture-agnostic, achieving 2.2x training step reduction on state space models.

Key finding: All existing MoE routing methods optimize for task quality and load balance. None incorporate real-time hardware energy signals into the routing decision itself.

3. Expert Behavior: What We Know

Clark et al. (2022) analyzed trained MoE language models and found **soft specialization**: experts develop token-type preferences (punctuation, entities, function words) but are not strictly partitioned. Routing entropy decreases in deeper layers, indicating more confident routing. This finding directly

informs our RQ2: tokens with low routing uncertainty (high confidence) are candidates for energy-aware redirection to low-power GPUs, as the quality impact of expert substitution is minimal when the router is already confident.

Complementing this, V-MoE (Riquelme 2021) found that vision experts learn **near-identical functions**, extracting common features rather than specializing by class. This expert interchangeability is precisely the property our energy penalty exploits: if experts are functionally similar, routing to a lower-power GPU's expert incurs negligible quality loss.

4. Energy Measurement: Tools & Gaps

The Green AI movement (Schwartz 2020) established the philosophical case for efficiency as a first-class metric. Patterson et al. (2021) provided empirical evidence: sparse MoE models consume less than 1/10th the energy of dense equivalents. Henderson et al. (2020) proposed a 13-metric reporting framework (experiment-impact-tracker), finding that 0% of NeurIPS 2019 papers reported carbon impact.

For measurement tooling, nvidia-smi/pynvml provides per-GPU power telemetry at ~1 Hz, though recent work (Burtscher 2023) cautions that A100/H100 GPUs only sample ~25% of runtime for power. PyJoules (De Vogeleeer 2022) offers a higher-level Python API with <5% overhead. Green Algorithms (Lannelongue 2021) provides a standardized decomposition: $\text{Carbon} = \text{Runtime} \times \text{Power} \times \text{PUE} \times \text{Carbon Intensity}$.

5. Comparison of Routing Strategies

Method	Top-k	Balance	Energy?	Key Result
Shazeer 2017	k=2-4	Importance + Load	No	24% lower PPL
Switch (Fedus 2022)	k=1	Simplified	No	7x speedup on C4
GShard (Lepikhin 2020)	k=2	Random dispatch	No	600B params, 100-lang MT
V-MoE (Riquelme 2021)	k=1-2	BPR	No	90.35% ImageNet
Soft MoE (Puigcerver 2023)	All (soft)	Implicit	No	No token dropping
Ours (proposed)	k=1, 2	Balance + Energy	YES	Pareto frontier

6. The Research Gap & Our Contribution

Our literature review reveals a clear structural gap at the intersection of MoE routing and energy measurement:

Gap: No existing work incorporates real-time, heterogeneous energy costs into the MoE routing decision function. Papers [1-6] build the routing machinery; papers [7-12] build the measurement machinery. Our project connects them: $\text{logits} = \mathbf{x} \cdot \mathbf{W}_g - \lambda \cdot \mathbf{e}_{\text{gpu}}$.

Our contribution is threefold: (1) a modified gating function with an additive energy penalty structurally analogous to Shazeer's exploration noise, (2) a multi-objective loss $\mathcal{L}_{\text{task}} + \alpha \cdot \mathcal{L}_{\text{balance}} + \beta \cdot \mathcal{L}_{\text{energy}}$, and (3) a comprehensive Pareto frontier analysis of quality vs. energy on C4 and GLUE.

7. Identified Baselines

Baseline 1 — Switch Transformer ($\lambda=0$): Standard energy-agnostic MoE using DeepSpeed-MoE. Represents the state-of-the-art sparse routing approach. Direct comparison on C4 validates our energy penalty's impact.

Baseline 2 — Random Router: Tokens uniformly randomly assigned to experts. Establishes a performance lower bound and helps debug the MoE container.

Baseline 3 — Dense Transformer: Standard Transformer with no MoE layers. Provides the compute-matched dense reference point from Patterson et al.'s energy analysis.

8. Risks & Mitigations for M1

Risk	Mitigation Applied
Overwhelmed by volume (15+ papers)	Triage reading accelerated by Claude for faster understanding: abstract/conclusion first, methods second, full read only if highly relevant. Claude was used as an assistant to explain dense passages in plain language, cross-check summaries, surface duplication between papers, and flag missing baselines. Result: all 15 papers reviewed within the allocated M1 window without loss of technical depth.
Missing recent un-indexed papers	Google Scholar alerts active for four queries: "mixture of experts routing", "energy-aware mixture of experts", "conditional computation transformer", "sparse transformer efficiency". Semantic Scholar citation alerts active on the three foundational papers (Shazeer 2017, Switch 2022, Patterson 2021). arXiv cs.LG/cs.CL/cs.DC firehose subscription intentionally deferred — two-signal triangulation (Scholar + Semantic Scholar) is sufficient coverage. Early validation: StructuredDNA (arXiv:2512.08968, Dec 2025) surfaced as potentially competing prior art (bio-physical energy-guided routing) and is flagged for urgent triage in M2 .
Confirmation bias in gap identification	Cross-validated the gap against comparison_table.xlsx with Claude as a second reader : explicitly checked every paper for energy-in-routing capability. The Gap-Analysis sheet confirms zero overlap with our contribution. Pending re-check against StructuredDNA once read.

References

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- [7] Schwartz et al., 2020. Green AI. Communications of the ACM.
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- [13] Hooker, 2021. The Hardware Lottery. Communications of the ACM.
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